

"Don't just plug and pray. Understand **WHAT** the data is telling you, **WHEN** to use which tool, and **HOW** to dodge the traps!"

1. 📊 DECISION MATRIX: What do I apply?

Goal: Compare Means (Continuous Data)

- 1 Sample vs Target (σ unknown) → **1-Sample t-test**
- 2 Independent Groups (A vs B) → **2-Sample t-test**
- Same Group Twice (Pre/Post test) → **Paired t-test**
- 3+ Independent Groups → **ANOVA (F-test)**

Goal: Categorical Data & Proportions

- Compare Win Rates / Percentages → **Z-test for Proportions**
- Does reality match expected ratio? → **Chi-Sq Goodness of Fit**
- Are 2 categorical vars linked? → **Chi-Sq Independence**

Goal: Relationships & Forecasting

- Predict continuous Y from X → **Linear Regression**
- Strength of linear trend (-1 to 1) → **Pearson Correlation (r)**
- Forecast flat time-series → **Moving Average / SES**
- Forecast data with a steady trend → **Holt's Method**
- Check if forecast errors are random → **White Noise Test**
- Soft clustering (overlapping groups) → **GMM & EM Algorithm**

3. 📊 MODULE 3: T-TESTS & CHI-SQUARE

t-tests (Use when Pop. Std Dev σ is UNKNOWN)

- **1-Sample:** $t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$ ($df = n - 1$)
- **2-Sample Indep:** $t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{1/n_1 + 1/n_2}}$ ($df = n_1 + n_2 - 2$)
Pooled Std Dev (s_p): $\sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$
- **Paired:** Subtract pairs first to get differences (d)! Treat it exactly like a 1-sample test. $t = \frac{\bar{d} - 0}{s_d / \sqrt{n}}$ ($df = n_{pairs} - 1$)

Chi-Square (χ^2) – Use for RAW COUNTS only!

Master Formula: $\chi^2 = \sum \frac{(O - E)^2}{E}$ (O=Observed, E=Expected)

- **Goodness of Fit (E):** Total Count $\times$ Expected %
- **Independence (E):** $\frac{\text{Row Total} \times \text{Col Total}}{\text{Grand Total}}$
- **Intuition:** If Observed (O) heavily deviates from Expected (E), χ^2 gets massive, meaning reality doesn't match theory (Reject H_0 !).

4. 📊 MODULE 4: Z-TESTS, C.I. & ANOVA

Z-tests for Proportions ($\hat{p} = \text{successes}/n$)

- **1-Sample:** $Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$
- **2-Sample:** $Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_{pool}(1 - \hat{p}_{pool})(1/n_1 + 1/n_2)}}$
Where $\hat{p}_{pool} = \frac{x_1 + x_2}{n_1 + n_2}$ (Total successes / Total attempts)

Confidence Intervals (CI)

Format: Point Estimate $\pm$ (Critical Value $\times$ Std Error)

- **Mean (σ known):** $\bar{x} \pm Z_{\alpha/2}(\sigma / \sqrt{n})$
- **Mean (σ unknown):** $\bar{x} \pm t_{\alpha/2, n-1}(s / \sqrt{n})$
- **Proportion:** $\hat{p} \pm Z_{\alpha/2} \sqrt{\hat{p}(1 - \hat{p})/n}$

One-Way ANOVA (3+ Groups) ($H_0 : \mu_1 = \mu_2 = \mu_3 \dots$)

- **Intuition:** Why "Analysis of Variance" to test means? If the variance *between* the groups is massive compared to the variance *within* the groups, the means must truly be different!
- $F = \frac{MSB}{MSW}$ ($MSB = \frac{SSB}{k-1}$, $MSW = \frac{SSW}{N-k}$)
- **Trap:** ANOVA only tells you **IF** a mean differs, not **WHICH** one.

2. 🎲 RULES OF THE GAME (Hypothesis)

- **Null (H_0):** The Status Quo. Always contains $=, \leq, \geq$. Assumes *no difference, no effect, or pure randomness*.
- **Alternate (H_a):** Your claim! Contains $\neq, <, >$.
- **1-Tail vs 2-Tail:**
 - "Did it *change*?" → 2-Tailed ($\neq$). **Divide α by 2!**
 - "Did it *increase/decrease*?" → 1-Tailed ($> / <$). Keep α .
- **The Golden Rule:** If $p\text{-value} \leq \alpha$, **Reject H_0** ("If P is low, the Null must go!"). Or Reject if $|\text{Test Stat}| > |\text{Crit Val}|$.

5. 📊 MODULE 5: REGRESSION & SES

Pearson Correlation (r): measures linear strength (-1 to 1).

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

Linear Regression ($\hat{y} = b_0 + b_1x$)

- **Slope (b_1):** $b_1 = r(\frac{s_y}{s_x})$. For every 1 unit increase in X , Y changes by b_1 .
- **Intercept (b_0):** $b_0 = \bar{y} - b_1\bar{x}$.

Simple Exponential Smoothing (SES)

Goal: Forecast flat data. Weights recent data more.

$$F_{t+1} = \alpha(\text{Actual } Y_t) + (1 - \alpha)(\text{Old Forecast } F_t)$$

- **The α parameter (0 to 1):** High α (e.g., 0.9) reacts quickly to recent shocks. Low α (e.g., 0.1) ignores noise and creates a smooth trend.

6. 📈 MODULE 6: FORECASTING & ML

Moving Average (MA): Avg of last k periods. Lags behind trends.

Holt's Method (Has Trend!)

Rule: MUST calculate in exact order: Level $\rightarrow$ Trend $\rightarrow$ Forecast.

1. **Level:** $L_t = \alpha Y_t + (1 - \alpha)(L_{t-1} + T_{t-1})$
2. **Trend:** $T_t = \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1}$
3. **Forecast:** $F_{t+m} = L_t + m \cdot T_t$ (for m periods ahead).

White Noise Test (Ljung-Box Test)

Checks if your forecast errors (residuals) are purely random.

- **THE BIG EXCEPTION:** This is the ONE test where you **WANT to Fail to Reject H_0** ($p > \alpha$). It means your model captured everything perfectly, leaving only random noise!

GMM & EM Algorithm

- **Intuition:** "Soft clustering". Instead of strict labels, points get *probabilities* of belonging to hidden normal distributions.
- **E-Step (Expectation):** Calculate "Responsibilities" (γ_{ic}). What is the prob that point i is in cluster c ? (Must sum to 1 for each point).
- **M-Step (Maximization):** Use those probabilities as weights to recalculate cluster Means (μ), Variances (Σ), and Weights (π).

7. ☠️ EXAM DAY TRAPS (Don't lose points!)

- **Variance vs Std Dev Trap:** Formulas need Std Dev (σ or s). If a prompt says "Variance is 25", do NOT square it! Square root it: $\sigma = \sqrt{25} = 5$.
- **The Chi-Square Percentage Trap:** χ^2 formulas ONLY work with raw counts (frequencies). Never plug % or decimals into $(O - E)^2 / E$.
- **SES Initialization:** If F_1 isn't given in a time-series question, standard convention is to assume $F_1 = Y_1$.
- **Standard Error vs Std Dev:** Remember, s is the spread of the *data*. $s/\sqrt{n}$ is the Standard Error (the spread of the *sample means*). Use Standard Error for hypothesis tests and CIs!